Dr. Tristan Bereau

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(Last updated: May 17, 2020)

Professional experience

University of Amsterdam, Amsterdam, Netherlands
- Assistant Professor, Van 't Hoff Institute for Molecular Sci. & Informatics Institute **2020–**

Max Planck Institute for Polymer Research, Theory Group, Mainz, Germany
- Emmy Noether (independent) group leader: *Importance Sampling in Chemical Space* **2016–2019**
- Group leader: Biomolecular simulations (Arbeitskreis Prof. Kurt Kremer) **2014–2016**

University of Basel and **Novartis**, Basel, Switzerland
- Postdoctoral assistant, Physical chemistry (Prof. Markus Meuwly) **2012–2013**

Education

Carnegie Mellon University, Pittsburgh, PA USA
- Ph.D., Physics (Prof. Markus Deserno) **2008–2011**

École Polytechnique Fédérale de Lausanne (EPFL), Lausanne, Switzerland
- B.Sc., Physics **2003–2006**

Secondary appointments

Guest Editor for *APL Materials* Special Issue **2020**
- Discovering Patterns in Disorder: Machine Learning for Fluctuating Mesoscopic Materials
- Together with Dr. Alpha Lee (U. Cambridge) and Prof. Daan Frenkel (U. Cambridge)

Editorial Board member, *Machine Learning: Science and Technology*, IOP Publishing **2019–**

Steering-committee member of the Carl-Zeiss-Stiftung on Algorithmic Intelligence **2019–2020**

Board member of the TRR146 **2018–2020**

Deputy member representative of FAIR-DI e.V. (FAIR Data Infrastructure) **2018–**
- Pillar C: "Soft-matter and biomolecular simulations"

Deputy member representative of the Max Planck Network "BiGmax" **2017–**
- Project Area 4: "Learning thermodynamic properties of materials"

Co-Head of the TRR146 Integrated research training group (IRTG) **2017–2020**

Junior PI at Max Planck Graduate Center (graduate school MPIP/JGU Mainz) **2017–2020**

Awards

"Hochschullehrer-Nachwuchs-Workshop 2019," Marl, Germany **09/2019**
- Invitation by GDCh Makromolekulare Chemie to selected young university lecturers

"Machine Learning for Physics and the Physics of Learning," CA USA **09-12/2019**
- Invited core participant of IPAM Long Program

Examination authorization (Prüfungsberechtigung), Physics, JGU Mainz **2018**

Emmy Noether Program, Deutsche Forschungsgemeinschaft (DFG) **2016**
- Independent research group
- Equivalence to Habilitation

Astrid and Bruce McWilliams Fellowship, CMU, Pittsburgh, PA USA **2010**
- 1-year full tuition and stipend scholarship

Student Travel Award – Biophysical Society Meeting, San Francisco, CA **2010**

Guy C. Berry Graduate Research Award – Mellon College of Science, CMU **2009**

Third-party funding

Carl-Zeiss-Stiftung "Algorithmic Intelligence as Emergent Phenomenon" **2019–2023**
- One of 4 core co-PIs (Consortium: **€3.0M**)
- Statistical predictability of physical systems (**€0.2M**)

Max Planck Network BiGmax on "Big Data in Materials Science" **2018-2022**
- Automated analysis of atom force probe detector maps (**€0.2M**)
- Learning thermodynamic properties of materials (**€0.2M**)

	Emmy Noether Program, Deutsche Forschungsgemeinschaft (DFG) • Independent research group (**€1.3M**) • Topic: Importance Sampling in Chemical Space	**2016–2020**
	TRR146 "Multiscale Modeling in Soft Matter"	**2014–2022**
	• A6: Dynamics in multiscale simulations (**€0.2M**)	**2014–2022**
	• B7: Machine learning in multiscale simulations (**€0.2M**)	**2018–2022**
Teaching (lectures)	*Advanced computer simulation techniques*, JGU Mainz • co-taught with Dr. Giovanni Settanni and Dr. Omar Valsson	**Spring 2019**
	Computer simulation techniques, JGU Mainz, Mainz, Germany • co-taught with Dr. Giovanni Settanni and Dr. Peter Virnau	**Fall 2018**
	Polymer physics and soft-matter theory, JGU Mainz, Mainz, Germany • co-taught with Prof. Burkhard Dünweg and Dr. Sara Jabbari-Farouji	**Fall 2018**
	Advanced computer simulation techniques, JGU Mainz, Mainz, Germany • co-taught with Dr. Denis Andrienko and Dr. Giovanni Settanni	**Spring 2018**
	Polymer physics and soft-matter theory, JGU Mainz, Mainz, Germany • co-taught with Prof. Burkhard Dünweg and Dr. Sara Jabbari-Farouji	**Fall 2017**
	Polymer physics and soft-matter theory, JGU Mainz, Mainz, Germany • co-taught with Prof. Burkhard Dünweg and Dr. Sara Jabbari-Farouji	**Fall 2016**
	Advanced computer simulation techniques, JGU Mainz, Mainz, Germany • co-taught with Dr. Denis Andrienko and Dr. Giovanni Settanni	**Spring 2016**
	Electronic structure calculations, University of Basel, Basel • co-taught with Prof. Markus Meuwly	**Fall 2012**
Teaching (guest lectures)	*Biomolecular simulations*, UvA, Amsterdam, Netherlands • Machine learning for coarse-grained simulations	**05/2020**
Teaching (summer & winter schools)	MolSim 2020, University of Amsterdam, Netherlands	**01/2020**
	CECAM: "Applied mathematics and machine learning perspectives on Big Data Problems in Computational Sciences," Mainz, Germany	**10/2019**
	CECAM/TRR School on Machine Learning, Mainz, Germany	**09/2018**
	CCCS School on machine learning in atomistic simulations, Vallico Sotto, Italy	**05/2017**
	CCCS School on coarse-graining, Engelberg, Switzerland	**03/2017**
	ESPResSo++/VOTCA tutorial, Schloss Waldthausen, Mainz, Germany	**10/2014**
Tutoring	CECAM/TRR School on Machine Learning, Mainz, Germany	**09/2018**
	CCCS School on coarse-graining, Engelberg, Switzerland	**03/2017**
	ESPResSo++/VOTCA tutorial, Schloss Waldthausen, Mainz, Germany	**10/2014**
	ESPResSo workshop, University of Stuttgart, Germany	**10/2010**
	Coarse-Grained Biomolecular Modeling workshop, Levi, Finland	**03/2010**
	Various teaching-assistant positions at both EPFL and CMU	**2006-2011**
Workshop organization	CECAM: "(Machine) learning how to coarse-grain," Mainz, Germany	**09/2020**
	CECAM: "Applied mathematics and machine learning perspectives on Big Data Problems in Computational Sciences," Mainz, Germany	**10/2019**
	Mainz Materials Simulation Days 2019, Mainz, Germany	**06/2019**
	International workshop for MaxNet on Big Data in Materials Science, Dresden, Germany	**04/2019**
	CECAM/TRR School on Machine Learning, Mainz, Germany (60 participants)	**09/2018**
	CECAM: "New frontiers in particle-based multiscale modeling," Mainz (60 participants)	**09/2018**
	Mainz Materials Simulation Days 2015, Mainz, Germany (100+ participants)	**06/2015**
	ESPResSo++/VOTCA tutorial, Schloss Waldthausen, Mainz, Germany	**10/2014**
Invited talks	MUST2020 NCCR Conference, Grindelwald, Switzerland postponed	**09/2020**
	CECAM "Non-covalent interactions in large molecules," Lausanne, Switzerland	**09/2020**
	PASC20 Conference, Geneva, Switzerland postponed	**07/2020**
	Conference on A FAIR Data Infrastructure for Materials Genomics virtual	**06/2020**

SIAM Conference "Mathematical Aspects of Materials Science," Bilbao, Spain postponed **05/2020**
Erice school "Free-energy landscapes," Erice, Italy postponed **05/2020**
Pompeu Fabra University of Barcelona, Barcelona, Spain cancelled **04/2020**
"Dresden Polymer Discussions," Meißen, Germany cancelled **03/2020**
CompBioMed "Machine learning meets modelling and simulation methods" virtual **03/2020**
- Recording online

PCoMS Seminar Series, Tohoku University, Sendai, Japan **02/2020**
Department of Chemical Engineering, Kyoto University, Kyoto, Japan **02/2020**
CANES Seminar Series, King's College London, London, United Kingdom **02/2020**
Physics@Veldhoven, Veldhoven, Netherlands **01/2020**
University of Marburg, Marburg, Germany **11/2019**
E-CAM "Building a molecular foundry" virtual **11/2019**
Leibniz Institute for Interactive Materials, Aachen, Germany **11/2019**
IPAM "Machine Learning for Physics and the Physics of Learning," UCLA, USA **10/2019**
- Recording online

CECAM "Beyond machine learning for quantum chemistry," Bremen, Germany **10/2019**
"Hochschullehrer-Nachwuchs-Workshop 2019," Marl, Germany **09/2019**
IPAM "Machine Learning for Physics and the Physics of Learning," UCLA, USA **09/2019**
- Recording online

American Chemical Society (ACS) Fall Meeting 2019, San Diego, CA USA **08/2019**
XXXI IUPAP Conference on Computational Physics (CCP2019), Hong Kong **07/2019**
"Advances in methods for multi-scale modelling," Leiden, Netherlands **06/2019**
"MolKin2019 Sampling, Design and Machine Learning," Berlin, Germany **06/2019**
IPAM "Many-Particle Systems with Machine Learning," Lake Arrowhead, USA **06/2019**
German Physical Society (DPG) meeting, Regensburg **03/2019**
American Physical Society (APS) meeting, Boston, USA **03/2019**
"Machine Learning and Reverse engineering for Soft Materials," Leiden, Netherlands **12/2018**
"Integrating Molecular Simulation with Machine Learning," Leiden, Netherlands **10/2018**
Max Planck Institute for Iron Research, Düsseldorf, Germany **08/2018**
"Modern Approaches to Coupling Scales In Materials Simulations," Lenggries, Germany **07/2018**
IPAM "Many-Particle Systems with Machine Learning," Lake Arrowhead, USA **06/2018**
Computational Chemistry Days, **Keynote lecture**, Helsinki, Finland **05/2018**
Theoretical Computational Chemistry Workshop, Engelberg, Switzerland **05/2018**
SimTech Conference, Stuttgart, Germany **03/2018**
German Physical Society (DPG) meeting, Cross-sectional symposium, Berlin, Germany **03/2018**
Department of Applied Mathematics, TU Eindhoven, Eindhoven, The Netherlands **02/2018**
Department of Chemistry, Ruhr University of Bochum, Bochum, Germany **01/2018**
German Chemical Society (GDCh), Satellite Meeting, Berlin, Germany **09/2017**
Symposium on Theoretical Chemistry (STC), University of Basel, Basel, Switzerland **08/2017**
Department of Physical Chemistry, University of Münster, Münster, Germany **07/2017**
Department of Physical Chemistry, University of Göttingen, Göttingen, Germany **06/2017**
IPAM "Many-Particle Systems with Machine Learning,", UCLA, USA **11/2016**
Department of Chemistry, Free University of Brussels, Brussels, Belgium **04/2016**
Department of Chemistry, University of Konstanz, Konstanz, Germany **04/2016**
"Transferability Issues in Multiscale Modelings," Mainz, Germany **12/2015**
Department of Chemistry, Freie Universität Berlin, Berlin, Germany **11/2015**
CCCS Symposium on Machine Learning, Basel, Switzerland **11/2015**
Statistical Physics and Low Dimensional Systems, Pont-à-Mousson, France **05/2015**
"Modeling Many-Body Interactions 2015," Lake Garda, Italy **05/2015**
CSP Workshop, University of Georgia, Athens GA USA **03/2015**
Faculty of Natural Sciences, University of Groningen, Groningen, The Netherlands **06/2015**
Mainz Materials Simulations Day (MMSD 2013), Mainz, Germany **06/2013**

Department of Chemistry, Penn State, State College, PA, USA	**05/2013**
Laboratoire de Biochimie Théorique, CNRS, Paris VII, France	**01/2013**
Forschungszentrum Jülich, Jülich, Germany	**11/2012**
Max Planck Institute for Polymer Research, Mainz, Germany	**11/2012**
Department of Biochemistry, University of Zürich, Switzerland	**12/2010**
Forschungszentrum Jülich, Jülich, Germany	**11/2010**
Max Planck Institute for Colloids and Interfaces, Potsdam, Germany	**07/2010**
INSERM, Paris, France	**06/2010**
Max Planck Institute for Polymer Research, Mainz, Germany	**07/2009**
Forschungszentrum Jülich, Jülich, Germany	**07/2009**

Supervision

University of Amsterdam, Amsterdam, Netherlands

- Bernadette Mohr, Ph.D. student — **2019–**

Max Planck Institute for Polymer Research, Theory Group, Mainz, Germany

- Yasemin Bozkurt Varolgüneş, Exchange Ph.D. student — **2018–2019**
- Martin Girard, Postdoc — **2018–**
- René Scheid, Master student — **2018–2019**
- Timon Wittenstein, Bachelor student — **2018–2019**
- Clemens Rauer, Postdoc — **2018–2019**
- Arghya Dutta, Postdoc — **2018–**
- Marc Stieffenhofer, Ph.D. student, **MPGC fellowship** — **2018–**
- Christian Hoffmann, Diplom student — **2018–2019**
- Bernadette Mohr, Diplom student — **2018–2019**
- Alessia Centi, Postdoc — **2017–2019**
- Kiran Kanekal, Ph.D. student, **MPGC fellowship** — **2016–**
- Roberto Menichetti, Postdoc — **2016–2018**
- Svenja Wörner, Ph.D. student — **2016–**
- Marius Bause, Ph.D. student, **MAINZ fellowship** — **2016–**
- Joseph F. Rudzinski, Postdoc, **Humboldt fellowship**, now Group leader — **2015–2019**
- Chan Liu, Ph.D. student — **2014–2019**

Doctoral defense committees

Yasemin Bozkurt Varolgüneş, Koç University, Istanbul, Turkey	**05/2020**
Chan Liu, Max Planck Institute for Polymer Research, Mainz, Germany	**10/2019**

Journal referee

Publons profile — **100+** reviews for **30+** international scientific journals

Peer-reviewed publications

Scholar profile — h-index: **18**; i10-index: **25** — (Open access: Preprint; Gold)

48. J. W. Barnett, C. R. Bilchak, Y. Wang, B. C. Benicewicz, L. A. Murdock, **T. Bereau**, S. K. Kumar, Designing exceptional gas-separation polymer membranes using machine learning, *Science Advances* **6** (2020)
    - Press releases on EurekAlert! and Phys.org

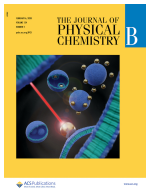
47. C. Scherer, R. Scheid, D. Andrienko, **T. Bereau**, Kernel-based machine learning for efficient simulations of molecular liquids, *J. Chem. Theory Comput.* **16** (2020)
    - LAMMPS custom implementation
    - VOTCA repository

46. Y. B. Varolgüneş, **T. Bereau**, J. F. Rudzinski, Interpretable embeddings from molecular simulations using Gaussian mixture variational autoencoders, *Mach. Learn.: Sci. Technol.* **1** (2020)

45. C. Hoffmann, A. Centi, R. Menichetti, **T. Bereau**, Molecular dynamics trajectories for 630 coarse-grained drug-membrane permeations, *Sci. Data* **7** (2020)
    - figshare database of 15,120 molecular dynamics trajectories

44. A. Centi, A. Dutta, S. H. Parekh, **T. Bereau**, Inserting small molecules across membrane mixtures: Insight from the potential of mean force, *Biophys. J.* **118** (2020)

43. A. H. R. Koch, S. Morsbach, **T. Bereau**, G. Lévêque, H. J. Butt, M. Deserno, K. Landfester, G. Fytas, Probing Nanoparticle-Membrane Interactions by Combining Amphiphilic Diblock Copolymer Assembly and Plasmonics, *J. Phys. Chem. B* **124** (2020)
    - Cover article



42. S. J. Wörner, **T. Bereau**, K. Kremer, J. F. Rudzinski, Direct route to reproducing pair distribution functions with coarse-grained models via transformed atomistic cross correlations, *J. Chem. Phys.* **151** (2019)

41. M. Bause, T. Wittenstein, K. Kremer, **T. Bereau**, Microscopic reweighting for nonequilibrium steady-state dynamics, *Phys. Rev. E Rapid Communications* **100** (2019)

40. K. H. Kanekal, **T. Bereau**, Resolution limit of data-driven coarse-grained models spanning chemical space, *J. Chem. Phys.* **151** (2019)

39. R. Menichetti and **T. Bereau**, Revisiting the Meyer-Overton rule for drug-membrane permeabilities, *Molecular Physics* **117** (2019)
    - Invited contribution to Special Issue *Liblice 2018*

38. C. Hoffmann, R. Menichetti, K. H. Kanekal, **T. Bereau**, Controlled exploration of chemical space by machine learning of coarse-grained representations, *Phys. Rev. E* **100** (2019)
    - Zenodo database of 1.3M drug-membrane transfer free energies

37. M. Girard, A. Ehlen, A. Shakya, **T. Bereau**, M. O. de la Cruz, Hoobas: A highly object-oriented builder for molecular dynamics, *Comput. Mater. Sci.* **167** (2019)

36. C. Schilling, T. Mack, S. Lickfett, S. Sieste, F. S. Ruggeri, T. Sneideris, A. Dutta, **T. Bereau**, R. Naraghi, D. Sinske, T. P. J. Knowles, C. V. Synatschke, T. Weil, B. Knöll, Sequence-Optimized Peptide Nanofibers as Growth Stimulators for Regeneration of Peripheral Neurons, *Adv. Funct. Mater.* **29** (2019)

35. R. Menichetti, K. H. Kanekal, **T. Bereau**, Drug-membrane permeability across chemical space, *ACS Central Sci.* **5** (2019)
    - Database of 500k permeability coefficients

34. J. F. Rudzinski, M. Radu, **T. Bereau**, Automated detection of many-particle solvation states for accurate characterizations of diffusion kinetics, *J. Chem. Phys.* **150** (2019)

33. **T. Bereau** and J. F. Rudzinski, Accurate structure-based coarse graining leads to consistent barrier-crossing dynamics, *Phys. Rev. Lett.* **121** (2018)
    - ESPResSo++ implementation

32. J. F. Rudzinski and **T. Bereau**, Structural-Kinetic-Thermodynamic Relationships Identified from Physics-based Molecular Simulation Models, *J. Chem. Phys.* **148** (2018)
    - Invited contribution to Special Issue *Markov Models of Molecular Kinetics*

31. C. Liu, K. Kremer, **T. Bereau**, Polymorphism of syndiotactic polystyrene crystals from multiscale simulations, *Adv. Theory Simul.* **1** (2018)

30. S. Morsbach, G. Gonella, V. Mailänder, S. Wegner, S. Wu, T. Weidner, R. Berger, K. Koynov, D. Vollmer, N. Encinas, S.-L. Kuan, **T. Bereau**, K. Kremer, T. Weil, M. Bonn, H.-J. Butt, K. Landfester, Engineering Proteins at Interfaces: From Complementary Characterization to Material Surfaces with Designed Functions, *Angewandte Chemie Int. Ed.* **57** (2018)

29. **T. Bereau**, R. DiStasio Jr., A. Tkatchenko, O. A. von Lilienfeld, Non-covalent interactions across organic and biological subsets of chemical space: physics-based potentials parametrized from machine learning, *J. Chem. Phys* **148** (2018)
    - Invited contribution to Special Issue *Data-Enabled Theoretical Chemistry*
    - Selected as Featured Article by the Editorial Board
    - Press release on AIP SciLight
    - Original implementation

28. J. F. Rudzinski and **T. Bereau**, The Role of Conformational Entropy in the Determination of Structural-Kinetic Relationships for Helix-Coil Transitions, *Computation* **6** (2018)

27. R. Menichetti, K. H. Kanekal, K. Kremer, **T. Bereau**, *In silico* screening of drug-membrane thermodynamics reveals linear relations between bulk partitioning and the potential of mean force, *J. Chem. Phys.* **147** (2017)

26. R. Menichetti, K. Kremer, **T. Bereau**, Efficient potential of mean force calculation from multiscale simulations: solute insertion in a lipid membrane, *Biochem. Biophys. Res. Commun.* **498** (2018)
    - Invited contribution to Special Issue *Multiscale Modeling*

25. J. Franz, **T. Bereau**, S. Pannwitt, V. Anbazhagan, A. Lehr, U. Nubbemeyer, U. Dietz, M. Bonn, T. Weidner, D. Schneider, Nitrated fatty acids modulate the physical properties of model membranes and the structure of transmembrane proteins, *Chem. Eur. J.* **23** (2017)

24. J. F. Rudzinski, **T. Bereau**, Concurrent parametrization against static and kinetic information leads to more robust coarse-grained force fields, *Eur. Phys. J. Special Topics* **225** (2016)
    - Invited contribution to Special Issue *Modern Simulation Approaches in Soft Matter Science: From Fundamental Understanding to Industrial Applications*

23. Q. Xiao, Y. Chen, **T. Bereau**, Y. Shi, An in-silico walker, *Chem. Phys. Lett.* **659** (2016)

22. **T. Bereau** and K. Kremer, Protein-Backbone Thermodynamics across the Membrane Interface, *J. Phys. Chem. B* **120** (2016)
    - Invited contribution to Special Issue *William M. Gelbart Festschrift*

21. K. El Hage, **T. Bereau**, S. Jakobsen, M. Meuwly, Impact of Quadrupolar Electrostatics on Atoms Adjacent to the Sigma-Hole in Condensed-Phase Simulations, *J. Chem. Theory Comput.* **12** (2016)

20. **T. Bereau**, D. Andrienko, and K. Kremer, Research Update: Computational materials discovery in soft matter, *APL Materials* **4** (2016)
    - Invited contribution to Special Issue *The Materials Genome Initiative*

19. J. F. Rudzinski, K. Kremer, **T. Bereau**, Consistent Interpretation of Molecular Simulation Kinetics using Markov State Models biased with External Information, *J. Chem. Phys. Communication* **144** (2016)

18. **T. Bereau**, W. F. D. Bennett, J. Pfaendtner, M. Deserno, M. Karttunen, Folding and Insertion Thermodynamics of the Transmembrane WALP Peptide, *J. Chem. Phys.* **143** (2015)
    - Invited contribution to Special Issue *Coarse graining of macromolecules, biopolymers, and membranes*

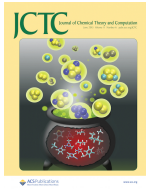
17. P. A. Cazade, H. Tran, **T. Bereau**, A. K. Das, F. Kläsi, P. Hamm, M. Meuwly, Solvation of fluoro-acetonitrile in water by 2D-IR spectroscopy: A combined experimental-computational study, *J. Chem. Phys.* **142** (2015)

16. **T. Bereau**, D. Andrienko, and A. von Lilienfeld, Transferable atomic multipole machine learning models for small organic molecules, *J. Chem. Theory Comput.* **11** (2015)

15. **T. Bereau** and M. Deserno, Enhanced sampling of coarse-grained transmembrane-peptide structure formation from hydrogen-bond replica exchange, *J. Membr. Biol.* **248** (2015)

14. **T. Bereau**, Better together: lipopeptide micelle formation enhances antimicrobial selectivity, *Biophys. J.* **109** (2015)
    - Invited contribution by Editors for *New and Notable*

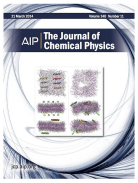
13. **T. Bereau** and K. Kremer, Automated Parametrization of the Coarse-Grained Martini Force Field for Small Organic Molecules, *J. Chem. Theory Comput.* **11** (2015)
    - Cover article
    - Original implementation
    - Implementation by MolSSI member Andrew Abi-Mansour



12. S. Jakobsen, **T. Bereau**, M. Meuwly, Multipolar force fields and their effects on solvent dynamics around simple solutes, *J. Phys. Chem. B* **119** (2015)

11. **T. Bereau** and O. A. von Lilienfeld, Toward transferable interatomic van der Waals interactions without electrons: The role of multipole electrostatics and many-body dispersion, *J. Chem. Phys.* **141** (2014)
    - Cover article

10. **T. Bereau**, Z.-J. Wang, and M. Deserno, More than the Sum of its Parts: Coarse-Grained Peptide-Lipid Interactions from a Simple Cross-Parametrization, *J. Chem. Phys.* **140** (2014)
    - Cover article



9. P. A. Cazade, **T. Bereau**, M. Meuwly, Computational Two-Dimensional Infrared Spectroscopy without Maps: N-Methylacetamide in Water, *J. Phys. Chem. B* **118** (2014)
    - Invited contribution to Special Issue *James L. Skinner Festschrift*

8. C. Kramer, **T. Bereau**, A. Spinn, K. R. Liedl, P. Gedeck, and M. Meuwly, Deriving Static Atomic Multipoles from the Electrostatic Potential, *J. Chem. Inf. Model.* **53** (2013)
    - Original implementation

7. **T. Bereau**, C. Kramer, and M. Meuwly, Leveraging Symmetries of Static Atomic Multipole Electrostatics in Molecular Dynamics Simulations, *J. Chem. Theory Comput.* **9** (2013)

6. **T. Bereau**, C. Kramer , F. W. Monnard, E. S. Nogueira, T. R. Ward, and M. Meuwly, Scoring Multipole Electrostatics in Condensed-Phase Atomistic Simulations, *J. Phys. Chem. B* **117** (2013)

5. **T. Bereau**, C. Globisch, C. Peter, and M. Deserno, Coarse-grained Simulations of the Salt-Stable Cowpea Chlorotic Mottle Virus subunit 26–49, *J. Chem. Theory Comput.* **8** (2012)

4. **T. Bereau**, M. Deserno, and M. Bachmann, Structural Basis of Folding Cooperativity in Model Proteins: Insights from a Microcanonical Perspective, *Biophys. J.* **100** (2011)

3. **T. Bereau**, M. Bachmann, and M. Deserno, Interplay between Secondary and Tertiary Structure Formation in Protein Folding Cooperativity, *J. Am. Chem. Soc.* **132** (2010)

2. **T. Bereau** and M. Deserno, Generic Coarse-grained Model for Protein Folding and Aggregation, *J. Chem. Phys.* **130** (2009)
    - ranked in June 2009's **Top 20 most downloaded** *J. Chem. Phys.* articles
    - ESPResSo++ implementation

1. **T. Bereau** and R. H. Swendsen, Optimized Convergence for Multiple Histogram Analysis, *J. Comput. Phys.* **228** (2009)
    - Original implementation

Proceedings

1. **T. Bereau**, Multi-timestep Integrator for the Modified Andersen Barostat, *Physics Procedia*, **68** (2015)

Book chapters

2. **T. Bereau**, Data-driven methods in multiscale modeling of soft matter, *Handbook of Materials Modeling. Volume 1 Methods: Theory and Modeling* (2018)

1. **T. Bereau** and M. Meuwly, Multipolar Force Fields for Atomistic Simulations, *Many-Body Effects and Electrostatics in Biomolecules* **233** (2016)